L3_INDEPENDENCE

L3_INDEPENDENCE REPORT

RESULTS: L3_INDEPENDENCE

PREREG.md sha256 5e9d28ff1785e76f3b8612edabf01c3f01df6e339e34304e86b1fb80c6336bcb. It was hashed after exploration (data/explore.sha256) and before the holdout run (scripts/test.py → data/test.json, logs/test.log).

Holdout: seeds ‘l3holdout-0..5999’ × 539 steps, with a 20-step burn-in. L3a uses every 8th step (n = 390,000). L3b uses all hits (346,349).

Check	Measured	Threshold	Verdict
A1 χ^2 indep. u mod 3 vs u mod 8	7.74 (df 10)	≤ 29.59	PASS
A2 fully specified (1/3)·w/32	χ^2 = 11.42 (df 17), TV = 0.0023	≤ 40.79, ≤ 0.006	PASS
A3 max conditional deviation from 1/3	0.0051	≤ 0.02	PASS
A4 χ^2 indep. u mod 3 vs u mod 64	94.95 (df 94)	≤ 142.12	PASS
A5 states with u ≡ 0 mod 4 after burn-in	0	= 0	PASS
L3b fibre law max abs deviation	$5.2 \cdot 10^{-4}$	≤ 0.005	PASS
L3b χ^2 on the six open fibres	3.94 (df 5)	≤ 20.52	PASS

Measured u mod 8 law (classes 1, 2, 3, 5, 6, 7): .2190, .2510, .1561, .0312, .2496, .0931. Predicted: .21875, .25, .15625, .03125, .25, .09375. The hit fibre law was (.0307, .2500, .0938, 0, .2184, .2505, .1567, 0), with zero hits in fibres 3 and 7 after burn-in.

L3a: PASS (Category B). L3b: PASS (Category B). Bridge: NO (residual-internal only). # NOTES: L3_INDEPENDENCE ## Lemmas - **(Category A, static, CRT).** For any c mod 2^k and a mod 3, the integers with u ≡ c (mod 2^k) and u ≡ a (mod 3) have density 1/(3·2^k). - The T3 branch is set by u mod 3, while u mod 2^k is set by low

binary digits. These are independent coordinates of Z/(3·2^k). - This is why the uniform-lift chain is natural. It says nothing about orbits. - **(Category A, from earlier work.)** Lemma V: after the opening run of pure ÷3 steps, u ≡/ 0 mod 4 for every later step. Confirmed again here, with 0 violations after burn-in. - **(Category B, upgraded evidence.)** Along real T3 orbits, u mod 3 is uniform and independent of u mod 8 and of u mod 64, within the pre-registered tolerance: - TV distance 0.0023 from (1/3) × law8. - Maximum conditional deviation 0.005, which is consistent with sampling noise. - The measured defect on the fibre law is ≤ $5.2 \cdot 10^{-4}$. - **L3 remains Category B.** No orbit-level proof exists. That would need equidistribution of the 3-adic digits that the map consumes, given the 2-adic history. The numbers are consistent with zero dependence. The bounded defect is < 0.0006 on each fibre and < 0.006 in TV.

Caveat

The thinning is every 8th step. Residual autocorrelation would only inflate χ^2 values, and all of them came out small.

Next residual-internal step

Try to make L3 exact in a skew-product form. Write T3 on $Z_2 \times Z_3$ as a 2-adic map driven by the 3-adic digit stream. Then test, or prove, that the branch sequence (u_i mod 3) is i.i.d.-uniform conditioned on past 2-adic states, e.g. by n-gram (block) tests of the branch word up to length 6 against 3^−L. If those tests pass, the fibre law becomes a theorem conditional on a single digit-normality hypothesis. Any Resonant-side fibre weighting still needs Arvin’s sign-off under the lock.

CORE_FREEZE respected: new folder only. No push or send. The order*/ folders and RTA_FOUNDATION_BOUND were not touched. C6, C10, C11 and E1–E3 were not touched, and no grading or Π was accepted.